Improving SAR Automatic Target Recognition via Trusted Knowledge Distillation From Simulated Data

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***Abstract*— In recent years, significant research has been conducted on utilizing simulated data to support synthetic aperture radar automatic target recognition (SAR-ATR) based on deep learning (DL) techniques. By distilling the dark knowledge extracted from simulated samples, the quality of the learned representations on measured samples can be effectively enhanced. However, our study highlights an important oversight in previous works: unquestioning trust in all simulated samples inevitably introduces the part of dark knowledge that is detrimental to SAR-ATR performance. To address this issue, we introduce evidential learning to estimate the confidence degree of the model after inputting simulated samples, thereby assessing the validity of the dark knowledge to be distilled. Then, the simulated-measured knowledge distillation (KD) process will be carried out in a trusted manner. Specifically, we encourage the model to prioritize distilling the dark knowledge with higher validity while avoiding the influence of inferior knowledge through a dynamic confidence weighting (DCW) method. Additionally, we transform the standard logits-based KD loss function into a feature-based one, giving the proposed method the ability to plug-and-play. The above aspects constitute the proposed trusted simulated-measured KD method for SAR-ATR. Multiple comparative studies on the simulated and measured paired labeled experiment (SAMPLE) dataset demonstrate the effectiveness of our proposed method, which not only achieves superior performance but also maintains the desired computational complexity in the inference phase.**

***Index Terms*— Automatic target recognition, deep learning (DL), knowledge distillation (KD), simulated data, synthetic aperture radar (SAR).**

I. Introduction

BENEFITING from the capability of synthetic aperture radar (SAR) system to capture images without being affected atmospheric factors, SAR automatic target recognition (SAR-ATR) has been widely applied in military reconnaissance, intelligence collection, target tracking, and other fields [1]. However, SAR images are often constrained by noise, clutter, and other interference factors due to complex imaging conditions. This poses a challenge for the development of high-performance SAR-ATR algorithms.

In the early stages of development, SAR-ATR primarily relied on hand-crafted feature extractors, with formalized and modelized expert knowledge at the core of such method design. Geometric structure features [2], electromagnetic-scattering features [3], transform features [4], and local invariant features [5] are commonly employed. Despite some progress, hand-crafted feature extractors have insufficient generalizability to SAR images due to the inherent limitations of their descriptors, making it difficult to achieve satisfactory SAR-ATR results [6]. To address this issue, deep learning (DL)-based SAR-ATR has been explored in recent years with fruitful results [7], [8], [9], [10], [11]. Unlike methods centered around expert knowledge, DL-based methods acquire dark knowledge in a data-driven manner, learning high-dimensional features from data to achieve remarkable SAR-ATR results. For example, Li et al. [12] proposed a multiscale deep convolutional learning model based on component analysis for SAR-ATR. Wang et al. [13] proposed a convolutional transformer for few-shot learning of SAR-ATR. Peng et al. [14] proposed a solution called contrastive feature alignment aiming to learn invariant representation for robust SAR-ATR.

In most previous DL-based studies, improved methodologies were proposed by decoupling the learning process of deep neural networks, which can be seen as leveraging the expert knowledge of network models to enhance the acquisition of data-driven dark knowledge. In contrast, it is more attractive to incorporate electromagnetic knowledge, an SAR-specific property, into DL-based SAR-ATR algorithms. However, directly embedding concrete electromagnetic knowledge into DL models is quite challenging due to conflicts with the data-driven core, which is an important reason why most of the studies have not addressed this aspect. Fortunately, the development of SAR image simulation with the capability for manual intervention has evolved into a mature technique over the years [15], [16], [17], [18]. This implies that electromagnetic knowledge can be indirectly acquired through learning on simulated data, which faithfully reproduces the

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electromagnetic scattering processes between electromagnetic waves and targets. Therefore, simulated data can serve as a valuable resource to supplement the requirements of DL models for electromagnetic knowledge.

Numerous studies have been conducted to explore the potential of simulated data in enhancing the SAR-ATR capability of DL models on measured data [19], [20], [21], [22], [23]. The core of the above-mentioned process is knowledge transfer based on pairs of simulated-measured target samples, which is generally accomplished through domain adaptation (DA) [24] or knowledge distillation (KD) [25] techniques. DA treats simulated data and measured data as two domains that share the same semantic information but exhibit distinct representation patterns. Knowledge transfer is achieved by aligning the feature representations of corresponding targets in both domains. In addition to DA, KD has emerged as a powerful tool for facilitating knowledge transfer among multiple views in recent years. Unlike the relatively independent feature extraction process in DA, KD employs the representation of one view, i.e., simulated, as the supervised signal to train another view, i.e., measured, to mimic its features or logits. In this process, informative dark knowledge, which is the hidden information that the model learns from the simulated data, is transferred to the model that is trained on the measured data. As a result, its knowledge transfer process exhibits greater generality and universality. However, there exists a challenge that cannot be resolved by current KD approaches:

Due to the difficulty in assessing the validity of the dark knowledge extracted from simulated data, there is no guarantee that every simulated sample will help the SAR-ATR model perform better on measured data.

During the learning process of all KD approaches for SAR-ATR, the dark knowledge from different simulated samples is given the same importance, with the underlying assumption that the quality or importance is basically stable for all simulated samples. However, in practice, SAR simulations will be affected by many factors such as modeling accuracy, background clutter, shadows, and data processing methods. Uneven quality of simulated data leads to differences in the validity of the dark knowledge extracted from them. Therefore, different simulated samples should be given different weights when they are used to support the learning of measured samples, thereby selectively distilling the dark knowledge that is most effective for improving performance. The current study ignores this issue, leading to unreliable KD. This inspires the concept of trusted simulated-measured KD.

In order to validate the feasibility of the above-mentioned KD approach, we conducted comprehensive motivating experiments on simulated and measured paired labeled experiment (SAMPLE) dataset [25]. It contains five identical models with different experimental settings. Method 1 entails pretraining a ResNet18 [26] model on simulated data and fine-tuning it with 20% of the measured data. Methods 2–4 use the same pretraining dataset as Method 1 but with 50% of the samples contaminated by Gaussian white noise of mean 0 and variances of 0.01, 0.05, and 0.1, respectively. This operation is aimed at exacerbating the uneven quality of simulated samples. The aforementioned four methods transfer knowledge from simulated data through a pretraining and fine-tuning approach. In contrast, Method 5 is designed to directly train on the measured data without pretraining. Hence, the comparison of Methods 1–4 with Method 5 is able to verify the validity of knowledge transfer from simulated data. In these experiments, training hyperparameters are kept consistent to ensure fairness. Experimental results are presented in Fig. 1.

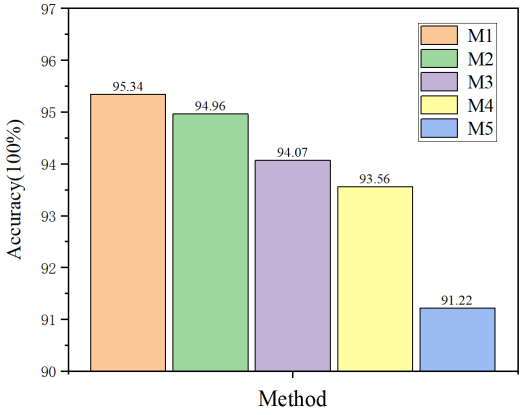


Fig. 1. Results of motivating experiments on SAMPLE dataset. M1: Pretraining on simulated data and fine-tuning on measured data. M2: Pretraining on simulated data with 50% $\mathcal{N}(0, 0.01)$ noise, and fine-tuning on measured data. M3: Pretraining on simulated data with 50% $\mathcal{N}(0, 0.05)$ noise, and fine-tuning on measured data. M4: Pretraining on simulated data with 50% $\mathcal{N}(0, 0.1)$ noise, and fine-tuning on measured data. M5: Training on measured data without pretraining.

The motivating experiments yield several significant conclusions. First, without transferring knowledge from simulated SAR images, M5 achieves an accuracy of 91.22%, the lowest of all methods. This indicates that the dark knowledge extracted from simulated data is quite useful for enhancing the SAR-ATR performance on measured data, even when the quality of some simulated data is slightly suboptimal. Second, as the variance of Gaussian white noise increases, the quality of the simulated images decreases, leading to a significant degradation in the accuracy of corresponding models on measured data. According to Fig. 1, the accuracy of M1–M4 is declined from 95.34% to 93.56%. This demonstrates that the simulated data with better quality can provide more useful dark knowledge.

Based on the aforementioned analysis, we believe that the key to achieving selective KD between simulated and measured data is how to assess the validity of dark knowledge. To address this issue, an intuitive approach is to classify different simulated samples with the softmax operator, and then use the predicted probabilities of the classification results as the assessments of their knowledge validity. However, the absence of uncertainty estimation in the softmax operator poses a challenge. Even when the prediction is incorrect, softmax assigns inflated predicted probabilities, impeding a reasonable analysis of the classification results. To obtain accurate validity assessments, it becomes imperative not only to determine the classification results but also to address inquiries such as "What is the confidence degree in the classification results?" and "What factors contribute to the high/low confidence degree?" This underscores the need for the development of novel mathematical tools that can independently estimate how confident the model is in its results.

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To make better use of simulated data to improve SAR-ATR, in this article, we first assess the quality of simulated samples by explicitly capturing the uncertainty in their classification results. Then the filtering of inferior dark knowledge is realized by assigning dynamic weights for each simulated sample. Furthermore, we empower the trusted KD with the plug-and-play capability for arbitrary DL-based SAR-ATR models by designing a novel distillation objective function that minimizes the similarity among paired simulated-measured features. In summary, the main contributions of this article are threefold.

1) We highlight a significant yet overlooked issue in current research studies on improving SAR-ATR with simulated data: not every simulated sample is equally effective in supporting better performance on measured data. Therefore, it is necessary to assess the validity of dark knowledge and let the model utilize the simulated data selectively for improving simulated-measured KD.
2) Following the pipeline of evidential learning, we estimate the confidence degree of simulated samples by modeling their output predicted probabilities as a Dirichlet distribution. Based on this, a confidence computing module (CCM) is designed to accurately assess the validity of the dark knowledge extracted from the simulated data.
3) We design a feature semantic enhancement module (FSEM) that leverages simulated samples to trustworthily enhance the semantic information in representations of measured samples. Within this module, we propose a novel KD learning objective, which not only adopts a dynamic confidence weighting (DCW) method to accept the confidence calculated by the CCM to help the model selectively receive the dark knowledge from different simulated samples, but also enables the plug-and-play integration of our proposed method into any DL-based SAR-ATR model.

The experimental results on SAMPLE dataset demonstrate the effectiveness of the proposed method.

II. Related Works

This section provides a comprehensive review of the research literature related to this article, including knowledge transfer from simulated data to measured data, KD, and uncertainty-based learning.

A. Knowledge Transfer From Simulated Data to Measured Data

DL algorithms, which rely on data-driven approaches, have demonstrated impressive performance in SAR-ATR tasks. However, the availability of labeled datasets containing an ample number of SAR measurement images at a significant scale is limited. As a result, there have been extensive research studies devoted to leveraging generated simulated data alongside measured data to enhance SAR-ATR performance, particularly under few-shot condition. Malmgren-Hansen et al. [24] showed the first study of transfer learning between a simulated dataset and a set of measured SAR images. Experiment results showed that convolutional neural networks (CNNs) pretrained on simulated data have significant advantages over CNNs trained only on measured data, especially when measured data are sparse. Lewis et al. [25] introduced the SAMPLE dataset, which consists of SAR images from the moving and stationary target acquisition and recognition (MSTAR) dataset and well-matched simulated images. Inkawhich et al. [27] focused on the case of having 100% simulated training data while testing on only measured data, which proposed data augmentation, model construction, loss functions, and ensemble techniques to enhance the representation learned from the simulated data. Ultimately, the results achieved over 95% accuracy on the SAMPLE dataset. To eliminate domain discrepancies between training and testing samples, He et al. [28] proposed a task-driven DA transfer learning method that can alleviate the degradation of recognition caused by the variance of depression angle between training and testing data. To address domain shift and task transfer problems caused by differences between simulated and measured data, Wang et al. [29] proposed a model that integrates meta-learning and adversarial DA. Cha et al. [30] introduced a method to refine simulated SAR data based on deep residual networks which learned a refinement function from simulated to measured data through a residual learning framework and used the function to refine simulated images. In Zhang et al. [31], a well-trained CNN was first utilized to preclassify all the test images. Then, the focus of the proposed method was to find the hard test samples that were easy to be misclassified according to the CNN classification confidence and reclassify them. Liu et al. [32] utilized cycle adversarial network, which could transfer simulated samples to be more similar to measured samples in the image domain. Sun et al. [33] explored the problems encountered when training fully on simulated data and testing on measured data and proposed a gradual DA recognition framework with pseudo-label denoising.

B. Knowledge Distillation

KD is an effective technique that has been widely used to transfer information from one network to another network [34], [35], [36], [37]. The concept was first defined by Buciluǎ et al. [38] and generalized by Hinton et al. [39]. After many years of development, numerous KD methods have been proposed with various intuitions. KD has been broadly applied to two distinct fields: model compression and knowledge transfer [40]. Ahn et al. [41] proposed an information-theoretic framework for knowledge transfer which formulates knowledge transfer as maximizing the mutual information between the teacher and the student networks. Peng et al. [42] proposed a new framework named correlation congruence for KD, which transferred not only the instance-level information but also the correlation between instances. Gou et al. [43] devised a collaborative KD via multiknowledge transfer that prompted both self-learning and collaborative learning in a unified framework. Zhang et al. [44] proposed a confidence-aware multiteacher KD method. The method adaptively assigned sample-wise reliability for each teacher prediction with the help of ground-truth labels, with those teacher predictions close to one-hot labels assigned large weights.

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In recent years, KD has been moving toward solving knowledge transfer between multimodal and multiview data and has achieved considerable research. Gupta et al. [45] used learned representations from a large labeled modality as a supervisory signal for training representations for a new unlabeled paired modality. Thoker and Gall [46] used pairs of sequences of both modalities as supervision, which were straightforward to acquire. Hoffman et al. [47] proposed a technique to adapt CNN-based object detectors trained on RGB images to effectively leverage depth images at test time to boost detection performance. Zhao et al. [48] generalized the distilled cross-modal knowledge learned from a source dataset, which contained paired examples from both modalities to the target dataset by modeling knowledge as priors on parameters of the student model. Kang et al. [49] presented a novel class incremental learning approach based on KD and provided a principled way to maintain the representations of old models while adjusting to new tasks effectively. Hong et al. [50] proposed the cross-modality KD network for monocular 3-D detection to efficiently and directly transfer the knowledge from LiDAR modality to image modality on both features and responses.

C. Uncertainty-Based Learning

As the use of DL algorithms becomes more and more widespread, confidence in neural network predictions becomes increasingly important. However, basic DL models cannot provide estimates of confidence. Therefore, many studies have been devoted to understanding and quantifying uncertainty in DL models [51]. Research studies based on Bayesian networks has made important contributions to the above topics, which estimate prediction uncertainty by approximating the moments of the posterior predictive distribution [52], [53]. Orthogonally to Bayesian neural nets that indirectly infer prediction uncertainty through weight uncertainties, Sensoy et al. [51] proposed explicit modeling of the using the theory of subjective logic. With the inspiration of evidential DL model, Han et al. [54] proposed a novel multiview classification algorithm, termed trusted multiview classification, providing a new paradigm for multiview learning by dynamically integrating different views at an evidence level. In Zou et al. [55], uncertainty was modeled explicitly using subjective logic theory, which treated the predictions of backbone neural network as subjective opinions by parameterizing the class probabilities of the segmentation as a Dirichlet distribution. Zhou et al. [56] developed a novel CNN integrating evidence theory to achieve fault classifications with prediction uncertainty estimation. The estimated prediction uncertainty could identify potential out-of-distribution samples.

III. Methodology

This section elaborates on our proposed framework for trusted simulated-measured KD. It consists of two network modules, i.e., the CCM and the FSEM. The core of CCM is to model the distribution of predicted probabilities with an evidence-parameterized Dirichlet distribution. This operation aims to estimate the confidence degree of the model for its output of the simulated data. Then we construct an association between classification confidence and dark knowledge, i.e., dark knowledge extracted from the simulated sample with higher classification confidence is more effective. Based on this, we devise the FSEM to implement the trusted distillation of dark knowledge. In its learning objective function, dynamic weights generated by DCW are used to represent the relative importance of different simulated samples, while the independent variable is designed to be the cosine similarity between simulated and measured features to ensure plug-and-play. Lastly, we present a comprehensive overview of the proposed method.

A. Confidence Computing Module

Softmax operator is commonly utilized to collect evidence that a sample belongs to a certain category in various DL models. For an input sample x, its feature representation $f(x)$, encoded by the network f, should be mapped through a fully connected (FC) layer c with K nodes (K is equal to the total number of categories) to obtain the following softmax-based predicted probability for the ith category:

$$\text{Prob}(i) = \frac{\exp\left(c^{(i)}(f(x))\right)}{\sum_{j=1}^{K}\exp\left(c^{(j)}(f(x))\right)}. \tag{1}$$

From the definition, it can be seen that each dimension of the predicted probability serves as evidence for x belonging to the corresponding category. However, since the ground truth values that predicted probabilities need to fit are one-hot encoded, the result of (1) is constrained to lie on a simplex, which limits the ability to capture the predicted uncertainty [57]. Consequently, incorrect predictions can also have sufficient evidence, i.e., high values of (1), making it hard to estimate the confidence of the output. In order to accurately collect evidence, second-order statistics, especially the variance, of predicted probabilities are needed, rather than a point estimation.

To address this issue, inspired by Sensoy et al. [51], we model the evidence and uncertainty through evidential learning. In this pipeline, predicted probabilities of the learned representations are modeled by their probability density function. Specifically, we note $\boldsymbol{\sigma} = (\text{Prob}(1), \ldots, \text{Prob}(K))$ as the predicted probability of an input sample $(x, \boldsymbol{y})$, where each $\text{Prob}(i) \in [0, 1]$ and $\sum_i \text{Prob}(i) = 1$, it should obey a multinomial distribution which describes the probability of K mutually exclusive events. Thus, the learning process with the softmax operator can be expressed as

$$\boldsymbol{y} \sim P(\boldsymbol{y}|\boldsymbol{\sigma}) = \text{Mult}(\boldsymbol{y}|\boldsymbol{\sigma}) \tag{2}$$

where $\text{Mult}(\cdot)$ is a multinomial mass function. As stated before, to introduce the variance of $\text{Mult}(\boldsymbol{y}|\boldsymbol{\sigma})$, we utilize its conjugate prior, i.e., the Dirichlet distribution, to model the predicted probabilities of the learned representations, which can be seen as a factory of multinomial distributions. Consequently, the learning objective shifts from modeling probability assignments for possible categories to modeling the density of such probability assignments, enabling the capture of second-order statistics. Correspondingly, the learning process should be transformed into the following formula:

$$\boldsymbol{\sigma} \sim P(\boldsymbol{\sigma}|x) = \text{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha}) \tag{3}$$

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where Dir(·) represents the evidence-parameterized Dirichlet distribution and $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$ is the parameter of $\mathrm{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha})$. The Dirichlet parameter is defined as: $\boldsymbol{\alpha} = \boldsymbol{e} + \mathbf{1}$, where $\boldsymbol{e} = (e_1, \ldots, e_K)$ represents the evidence collected by the DL model for an input sample being classified into a certain category, and obtained by the following ReLu operator:

$$\boldsymbol{e} = \max(c(f(x)), 0). \tag{4}$$

The above operator ensures that the evidence collected for each category is represented by a positive value. Then the belief b_i for the ith category and uncertainty u can be expressed as

$$b_i = \frac{e_i}{S} = \frac{\alpha_i - 1}{S}, \quad u = \frac{K}{S} \tag{5}$$

where $S = \sum_i (e_i + 1) = \sum_i \alpha_i$ is the Dirichlet strength. Thus, the model can express "*I do not know*" as an opinion for the truth over possible states. With the aforementioned definitions, the probability density function for possible values of the probability mass function $\boldsymbol{\sigma}$ can be formulated as

$$\mathrm{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha}) = \begin{cases} \dfrac{1}{B(\boldsymbol{\alpha})} \displaystyle\prod_{i=1}^{K} \sigma_i^{\alpha_i - 1} & \text{for } \boldsymbol{\sigma} \in \mathcal{S}_K, \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

where $B(\boldsymbol{\alpha})$ is the K-dimensional multinomial beta function, and $\mathcal{S}_K$ is the K-dimensional unit simplex

$$\mathcal{S}_K = \left\{ \boldsymbol{\sigma} \middle| \sum_{i=1}^{K} \sigma_i = 1 \text{ and } 0 \le \sigma_1, \sigma_2, \ldots, \sigma_K \le 1 \right\}. \tag{7}$$

Thanks to the conjugate prior relationship between $\mathrm{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha})$ and $\mathrm{Mult}(\boldsymbol{y}|\boldsymbol{\sigma})$, by making a simple modification of the classical cross-entropy loss function, the following adjusted cross-entropy loss [51] can be obtained to implement the learning of (3)

$$\begin{aligned} &\mathbb{E}_{\boldsymbol{\sigma}_i \sim \mathrm{Dir}(\boldsymbol{\sigma}_i|\boldsymbol{\alpha}_i)}\left[\log P\left(\boldsymbol{y}_i|\boldsymbol{\sigma}_i\right)\right] \\ &= \int \mathcal{L}_{\mathrm{CE}} \cdot \mathrm{Dir}(\boldsymbol{\sigma}_i|\boldsymbol{\alpha}_i) d\boldsymbol{\alpha}_i \\ &= \int \left[-\sum_{j=1}^{K} y_{ij} \log\left(\sigma_{ij}\right) \right] \frac{1}{B(\boldsymbol{\alpha}_i)} \prod_{j=1}^{K} \sigma_{ij}^{\alpha_{ij}-1} d\boldsymbol{\sigma}_i \\ &= \sum_{j=1}^{K} y_{ij} \left(\psi(S_i) - \psi\left(\alpha_{ij}\right)\right) \end{aligned} \tag{8}$$

where $\psi(\cdot)$ is the digamma function, $\mathcal{L}_{\mathrm{CE}}$ denotes the cross-entropy loss. In the context of multiclass classification, the resulting Dirichlet distribution, after the optimization of (8), should be concentrated on the vertex corresponding to the ground truth label $\boldsymbol{y}$ in the simplex. This implies that the parameter $\boldsymbol{\alpha}$ of the Dirichlet distribution, except for the dimension corresponding to the correct label, should be as close to 1 as possible. Typical examples are visualized in Fig. 2, where the desired situation is shown in Fig. 2(b).

However, (8) does not place any constraint on the samples that cannot be correctly classified. In this case, we tend to reduce all evidence to zero, where the $\mathrm{Dir}(\boldsymbol{\sigma}|\boldsymbol{\alpha})$ corresponds to a uniform distribution, and the uncertainty is 1. This can be achieved by adding the following loss term based on Kullback-Leibler (KL) divergence:

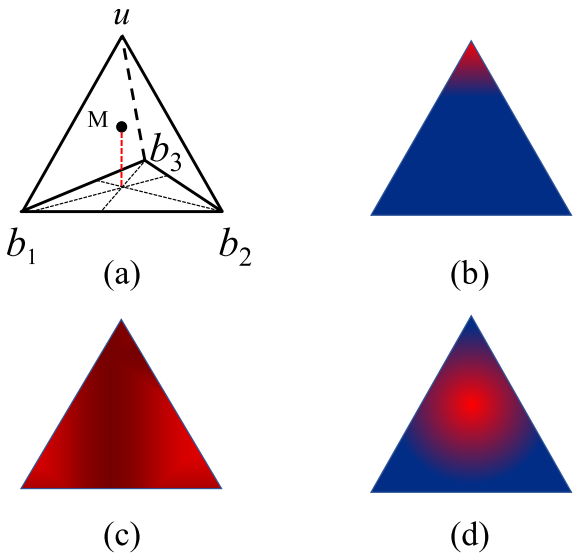


Fig. 2. Typical examples of the Dirichlet distribution for a triple classification task. (a) Opinion $\mathrm{M} = (\boldsymbol{b}, u)$ is visualized as a point inside a tetrahedron. (b) Confident prediction. (c) None evidence. (d) Conflicting evidence. Colors from blue to red indicate values from low to high.

$$\begin{aligned} &D_{\mathrm{KL}}\left[\mathrm{Dir}\left(\boldsymbol{\sigma}_i|\tilde{\boldsymbol{\alpha}}_i\right)||\mathrm{Dir}(\boldsymbol{\sigma}_i|[1, 1, \ldots, 1])\right] \\ &\quad = \log\left(\frac{\Gamma\left(\sum_{k=1}^{K} \tilde{\alpha}_{ik}\right)}{\Gamma(K) \prod_{k=1}^{K} \Gamma\left(\tilde{\alpha}_{ik}\right)} \right) \\ &\qquad + \sum_{i=1}^{K} \left(\tilde{\alpha}_{ik} - 1\right) \left[\psi\left(\tilde{\alpha}_{ik}\right) - \psi\left(\sum_{j=1}^{K} \tilde{\alpha}_{ij} \right) \right] \end{aligned} \tag{9}$$

where $\tilde{\boldsymbol{\alpha}}_i = \boldsymbol{y}_i + (\mathbf{1} - \boldsymbol{y}_i) \odot \boldsymbol{\alpha}_i$, and $\Gamma(\cdot)$ is the gamma function. Based on the above analysis, the overall loss function for optimizing the model to collect evidence can be written as follows:

$$\begin{aligned} \mathcal{L}_{\mathrm{CCM}}\left(x_i, \boldsymbol{y}_i\right) = &\mathbb{E}_{\boldsymbol{\sigma}_i \sim \mathrm{Dir}(\boldsymbol{\sigma}_i|\boldsymbol{\alpha}_i)}\left[\log P\left(\boldsymbol{y}_i|\boldsymbol{\sigma}_i\right)\right] \\ &- \lambda_t D_{\mathrm{KL}}\left[\mathrm{Dir}\left(\boldsymbol{\sigma}_i|\tilde{\boldsymbol{\alpha}}_i\right)||\mathrm{Dir}(\boldsymbol{\sigma}_i|[1, 1, \ldots, 1])\right] \end{aligned} \tag{10}$$

where hyperparameter λ_t serves to balance the loss of correctly and incorrectly classified samples. As the training proceeds, λ_t is gradually adjusted to prevent the model from excessively focusing on distribution disparities in the early stages. By optimizing $\mathcal{L}_{\mathrm{CCM}}$, we can model the output as an evidence-parameterized Dirichlet distribution and use it to learn feature representations $f(x)$, confidence degree (belief mass $\boldsymbol{b}$ in (5) and uncertainty u). In this way, for input simulated samples $(x^{v_1}, \boldsymbol{y})$, the CCM can be constructed by optimizing $\mathcal{L}_{\mathrm{CCM}}(x^{v_1}, \boldsymbol{y})$, which not only learns effective representations from simulated data but also computes the confidence of the extracted dark knowledge for each simulated sample.

B. Feature Semantic Enhancement Module

The CCM designed in Section III-A modifies the conventional classification operation in order to estimate the

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confidence of simulated data. This resulted in the widely used logits-based KD loss function no longer being applicable. To address this issue, in this module, we devise a novel KD loss function to implement the trusted distillation of knowledge from simulated data to measured data.

Consistent with the previous definitions, we denote the measured data as $(x^{v_2}, \boldsymbol{y})$, thus the training set can be expressed as: $\{X, Y\} = \{(x_1^{v_1}, x_1^{v_2}, \boldsymbol{y}_1), \ldots, (x_N^{v_1}, x_N^{v_2}, \boldsymbol{y}_N)\}$. In conventional learning models with the encoders f_1, f_2 and FC layers c_1, c_2, the logits-based KD loss can be formulated as follows:

$$\begin{aligned} \text{Loss} &= \sum_{i=1}^{N} D_{\text{KL}}\left[p\left(x_i^{v_1}, \tau\right) || p\left(x_i^{v_2}, \tau\right)\right], \quad \text{where} \\ p\left(x_i^{v_m}, \tau\right) &= \text{softmax}\left(c_m\left(f_m\left(x_i^{v_m}\right)\right)/\tau\right), \quad m = 1, 2 \end{aligned} \tag{11}$$

where temperature coefficient τ is utilized to regularize the smoothness of the output probability distribution, and the definition of softmax$(\cdot)$ can be found in (1). By minimizing the KL divergence between the softmax-activated output logits of the two models [38], [39], informative dark knowledge can be transferred from the model trained on simulated data to the model for measured data.

However, on the one hand, outputs of the CCM are no longer modeled as softmax-activated logits, and on the other hand, (11) may not effectively capture the underlying feature structures and patterns [58]. Consequently, introducing feature-level information to achieve trusted KD is essential for performance enhancement and ensuring plug-and-play capability. Under this consideration, we construct the following cosine similarity-based metric to evaluate how well the measured features mimic the simulated features:

$$\begin{aligned} \text{Sim}\left(x_i^{v_1}, x_i^{v_2}\right) &= \frac{\boldsymbol{z}_i^{v_1} \cdot \boldsymbol{z}_i^{v_2}}{\|\boldsymbol{z}_i^{v_1}\| \|\boldsymbol{z}_i^{v_2}\|} \\ &= \frac{\sum_j z_{i,j}^{v_1} z_{i,j}^{v_2}}{\sqrt{\sum_j \left(z_{i,j}^{v_1}\right)^2} \sqrt{\sum_j \left(z_{i,j}^{v_2}\right)^2}} \end{aligned} \tag{12}$$

where $\boldsymbol{z}_i^{v_1}$ and $\boldsymbol{z}_i^{v_2}$ denote the feature maps extracted by encoders f_1 and f_2 from the ith simulated-measured sample pair, i.e., $\boldsymbol{z}_i^{v_m} = f_m(x_i^{v_m})$, $m = 1, 2$, and j is the index of feature dimension. It can be seen from the definition that values of $\text{Sim}(x^{v_1}, x^{v_2})$ range between -1 (representing perfect anti-correlation) and 1 (representing perfect correlation) [59].

Our objective is to maximize the shared semantic information in the two feature maps with respect to what they exhibit in each dimension, while making the transferred dark knowledge as trustworthy as possible. However, due to the inherent differences between simulated and measured samples, not all the simulated data can effectively assist the learning on measured data. Therefore, treating all simulated samples equally during the distillation process is not reasonable. Fortunately, we can utilize the confidence degree calculated by the CCM to evaluate the validity of the dark knowledge. Intuitively, the target semantic information obtained from simulated samples with high confidence is more reliable than that with low confidence. Therefore, we believe that sample pairs corresponding to the simulated samples with high confidence should be aligned as much as possible. To achieve this, the following KD loss function is induced based on reweighting the importance of the dark knowledge extracted from each simulated sample:

$$\mathcal{L}_{\text{FSEM}}(x^{v_1}, x^{v_2}, \omega) = \sum_{i=1}^{N} w_i \text{Sim}\left(x_i^{v_1}, x_i^{v_2}\right) \tag{13}$$

where w_i denotes the weight of ith simulated sample, which is defined as follows:

$$w_i = \frac{1/u_i}{\sum_{j=1}^{N} 1/u_j}. \tag{14}$$

It can be seen from (14) that the higher the uncertainty of a simulated sample, the lower the importance it is assigned in the distillation process. Through the application of the aforementioned weights, we can achieve the dynamic reweighting of different simulated samples based on their confidence, which is our proposed DCW method. This prompts the model to adaptively reduce its attention to low-quality dark knowledge, preventing it from affecting the classification performance on measured samples.

C. Overall Learning Objective

Based on the aforementioned CCM and FSEM, we establish a plug-and-play learning framework, called trusted simulated-measured KD, to fulfill the task of leveraging simulated data to enhance the SAR-ATR performance. The unified framework for this approach is depicted in Fig. 3.

The overall training process of the proposed method is summarized as follows: After encoding the input simulated samples with the encoder f_1 and the FC layer c_1, we can obtain the parameter α of the Dirichlet distribution (6). According to (5), the belief mass and uncertainty for each simulated sample can also be calculated using the CCM. Then the KD from simulated samples to measured samples can be achieved through the FSEM. Measured samples in the training mini-batch should first be encoded by f_2. Then similarities of paired feature maps should be computed according to (12). Next, we need to calculate the weights of each simulated sample based on (14) and reweight the KD loss for different sample pairs based on their confidence degrees. Finally, the FC layer c_2 and a softmax-based classification operation are used for prediction. Hence, the overall learning objective function of the proposed trusted simulated-measured KD framework can be expressed as follows:

$$\begin{aligned} \mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CE}}(x^{v_2}, \boldsymbol{y}) + \lambda_1 \mathcal{L}_{\text{CCM}}(x^{v_1}, \boldsymbol{y}) \\ + \lambda_2 \mathcal{L}_{\text{FSEM}}(x^{v_1}, x^{v_2}, \omega) \end{aligned} \tag{15}$$

where hyperparameters λ_1 and λ_2 control the contributions of CCM and FSEM to the overall learning objective. From the definition of (15), it can be seen that $\mathcal{L}_{\text{CCM}}$ is the loss of learning feature representations and confidence degrees from simulated samples, $\mathcal{L}_{\text{FSEM}}$ is the loss of implementing feature-level KD through simulated-measured sample pairs and the computed confidence degrees, and $\mathcal{L}_{\text{CE}}$ is the loss of learning feature representations from measured samples. In conventional DL-based SAR-ATR models, only the first item of (15) exists. According to the definition, the second

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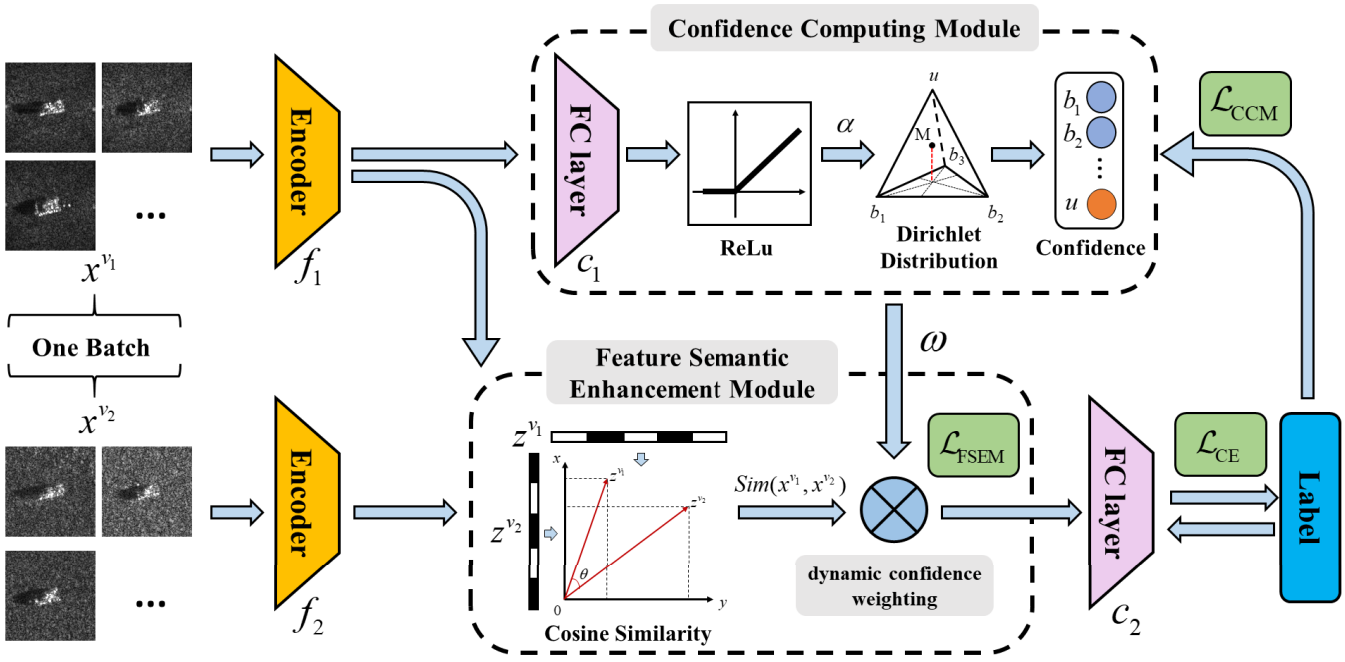


Fig. 3. Illustration of the proposed trusted simulated-measured KD method. It takes paired simulated-measured samples (x^{v_1}, x^{v_2}) as the input and obtains feature representations z^{v_1}, z^{v_2} by encoders f_1, f_2, respectively. The CCM uses ground truth labels and the evidence-parameterized Dirichlet distribution to estimate the confidence degree of z^{v_1}. Meanwhile, the FSEM enhances the learned features z^{v_2} by distilling the informative dark knowledge from z^{v_1}. For different sample pairs in a training mini-batch, the uncertainty calculated by the CCM is used for the DCW method to obtain the reweighted KD loss $\mathcal{L}_{\text{FSEM}}$. Besides, the cross-entropy loss $\mathcal{L}_{\text{CE}}$ is used to enable f_2 to receive supervision information from ground truth labels. Finally, $\mathcal{L}_{\text{CCM}}$, $\mathcal{L}_{\text{FSEM}}$, and $\mathcal{L}_{\text{CE}}$ constitute the overall loss function $\mathcal{L}_{\text{Total}}$ of the model.

and third terms of (15), which represents the process of trusted simulated-measured KD, can be seamlessly inserted into any model, reflecting the plug-and-play capacity of the proposed method. In summary, we outline the overall training process as Algorithm 1.

Once the training is finished, the modules used for distilling knowledge from simulated data will be discarded. Only the components responsible for feature extraction and classification on measured data will be retained. Therefore, the proposed method is not only plug-and-play during the training phase, but also does not introduce any additional computational complexity during the testing phase.

IV. Experiments

In this section, we first introduce the dataset used in the experiments. Then we proceed to demonstrate the performance of both the proposed method and the compared methods. Finally, we analyze the effect of the hyperparameters in the proposed method on its target recognition performance.

A. SAMPLE Dataset

The publicly available MSTAR dataset [60] has been a valuable resource for the development of SAR-ATR over the past two decades, resulting in impressive target recognition outcomes. However, due to the vast number of possible sensor parameters, target configurations, and environmental conditions, it is impractical to collect a sufficient amount of measured data that covers all imaging conditions. Consequently, simulated data must be generated to augment the available measured dataset. In this regard, we utilize the SAMPLE dataset [25], which comprises measured SAR images from the MSTAR dataset along with well-matched simulated data, as illustrated in Fig. 4. By aligning target configurations and sensor parameters between the measured and simulated data, the SAMPLE dataset offers a suitable platform for investigating the utilization of measured and simulated SAR images.

The SAMPLE dataset leverages asymptotic ray tracing to predict the electromagnetic characteristics of targets. To achieve this, each target is meticulously modeled using computer-aided design techniques, with material properties assigned to each surface based on annotations collected by MSTAR and intelligent reasoning about component composition. Each material is assigned an appropriate electromagnetic property value. During the simulation process, an electromagnetic solver takes these values into account when calculating radar echoes from the target. In order to simulate each target, it is positioned atop a ground surface characterized by random roughness, which provides a background for each image. This approach ensures that each simulated SAR image exhibits a background with speckles.

It is important to highlight that during the simulation of the SAMPLE dataset, imaging parameters identical to those employed in the acquisition of measured images in the MSTAR dataset are utilized. These imaging parameters are presented in Table I. Additionally, the SAMPLE dataset offers pairwise alignment between the simulated data and measured data, ensuring the consistency of target semantic information.

B. Experimental Settings

We validate the effectiveness of our proposed method by testing the performance of models based on five different

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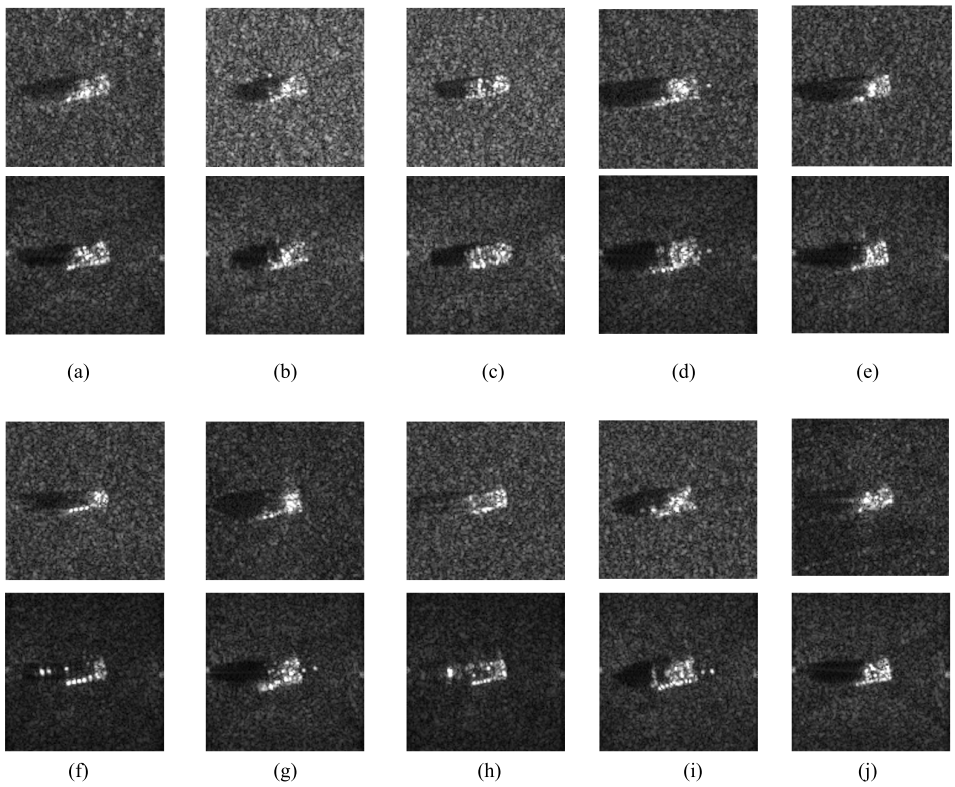


Fig. 4. Ten types of military targets in the SAMPLE dataset. Measured images are on the top row and simulated images are on the bottom row. (a) 2S1. (b) BMP2. (c) BTR70. (d) M1. (e) M2. (f) M35. (g) M648. (h) M60. (i) T72. (j) ZSU23.

TABLE I
TYPICAL PARAMETERS OF SIMULATION

Parameter	Value
Range resolution	0.3 m
Range pixel spacing	0.2 m
Range extent	25.8 m
Cross-range resolution	0.3 m
Cross-range pixel spacing	0.2 m
Cross-range extent	25.8 m
Bandwidth	591 MHz
Center frequency	9.6 GHz
Image size	128×128
Polarization	HH
Elevations	$14^\circ - 18^\circ$
Taylor weighting	-35 dB

backbones, including VGG16 [61], ResNet18, ResNet34 [26], SENet [62], and ResNeXt34 [63]. The above-mentioned backbones are all well-established and widely recognized DL models, frequently employed as benchmarks in SAR-ATR researches. In addition, we test several DA and KD models based on these backbones to compare them with the proposed method.

As for DA methods used for performance comparison, we select deep adaptation network (DAN) [64], deep correlation alignment (Deep Coral) [65], domain adversarial neural network (DANN) [66], conditional domain adversarial network (CDAN) [67], and deep joint distribution adaptation network (DJDAN) [68]. These methods have demonstrated remarkable performance in DA researches within SAR field. As for KD methods used for performance comparison, we select Vanilla KD [39], activation boundary (AB) [69], Fitnet [70], neural selective transfer (NST) [71], similarity preservation (SP) [72], and Probabilistic Knowledge Transfer (PKT) [73]. All the above comparing methods are tested based on the involved five backbones.

As for the experimental setup, we follow the approach outlined in Inkawhich et al. [27], which divides the SAMPLE dataset into two scenarios, referred to as "Training Scenario I" and "Training Scenario II." In the Training Scenario I, we select 20% of the simulated and measured sample pairs from the SAMPLE dataset as the training dataset and utilize the remaining 80% of measured samples as the testing dataset for our experiments, without considering the difference of depression angles. In the Training Scenario II, we take the sample pairs with depression angles ranging from 14° to 16° as the training set, while the measured samples with depression angle of 17° as the testing set. Then we select 20% and 50% sample pairs from the training set for training, and test on all the testing data. The specific experiment sample settings of the above two scenarios are shown in Tables II and III.

In our experiments, we intentionally refrain from applying sophisticated data preprocessing methods, such as cropping, rotation, and despeckling, which are commonly employed in practical SAR-ATR tasks to further enhance the performance of target recognition models [8], [9], [10]. This serves to better demonstrate the applicability and robustness of the model itself in SAR-ATR.

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Algorithm 1 Training Process of the Trusted Simulated-Measured Knowledge Distillation Method

Require:

Initializing model parameters W in all network modules; Setting the numbers of epoch I, mini-batch size B, learning rate η, hyperparameters λ_1, λ_2, and λ_t; Preparing the training and validation datasets.

for iteration in I: **do**
  for samples in mini-batch: **do**
    Forward propagation of $(x_i^{v_1}, x_i^{v_2}, \boldsymbol{y}_i)$ based on trainable parameters in f_1, f_2 and c_1, c_2
    # obtain the parameter of the Dirichlet distribution according to (4)-(5)
    $\boldsymbol{e}_i \leftarrow \max(c_1(f_1(x_i^{v_1})), 0)$; $\boldsymbol{\alpha}_i \leftarrow \boldsymbol{e}_i + \mathbf{1}$
    # obtain the uncertainty u_i according to (5) and $\boldsymbol{e}_i$
    $u_i \leftarrow K/(\sum_k (e_k + 1))$
    # obtain the $\mathcal{L}_{\mathrm{CCM}}$ according to (10) and $\boldsymbol{\alpha}_i$
    $\mathcal{L}_{\mathrm{CCM}} \leftarrow \mathcal{L}_{\mathrm{CCM}}(x_i^{v_1}, \boldsymbol{y}_i)$
    # obtain the similarity Sim according to (12)
    $Sim \leftarrow Sim(x_i^{v_1}, x_i^{v_2})$
    # obtain the weight ω according to (14) and u_i
    $\omega_i \leftarrow (1/u_i)/(\sum_j 1/u_j)$
    # obtain the $\mathcal{L}_{\mathrm{FSEM}}$ according to (13), ω_i, and Sim
    $\mathcal{L}_{\mathrm{FSEM}} \leftarrow \mathcal{L}_{\mathrm{FSEM}}(x_i^{v_1}, x_i^{v_2}, \omega_i)$
    # obtain $\mathcal{L}_{\mathrm{CE}}$ via $(x_i^{v_2}, \boldsymbol{y}_i)$
    # obtain $\mathcal{L}_{\mathrm{Total}}$ according to (15)
    $\mathcal{L}_{\mathrm{Total}} \leftarrow \mathcal{L}_{\mathrm{CE}} + \lambda_1 \mathcal{L}_{\mathrm{CCM}} + \lambda_2 \mathcal{L}_{\mathrm{FSEM}}$
    # update model parameters through back propagation
    $W \leftarrow W - \eta \nabla_W \mathcal{L}_{\mathrm{Total}}$
  end for
end for

Ensure:

Saving f_2, c_2 for the model with the highest validation accuracy.

Under different experimental settings, we introduce the following two baselines to better illustrate the effectiveness of distilling trusted knowledge from simulated data. The first baseline, referred to as "Baseline I (Scratch)," involves training the respective models solely on measured data without the use of simulated data. The second baseline directly mixes the measured and simulated samples as its training set, referred to as "Baseline II (Mix)."

The mini-batch stochastic gradient descent-based method is used for model optimization, with a mini-batch size of 15, and a training epoch of 100. In order to accelerate the optimization process and obtain more precise approximate solutions, the Adam optimizer is employed with a learning rate of 0.01. Moreover, the λ_t in (10) is set to $\min(1.0, \mathrm{epoch}/15)$, and the hyperparameters λ_1 and λ_2 in (15) are both set to 0.1. The experiments of this study employ an NVIDIA RTX A6000 GPU, and the program is implemented using the PyTorch DL framework.

C. Experiment Results and Analysis

1) Experiment Results Under Training Scenario I: During experiments, we employ the pertinent algorithms to address the ten-class target recognition problem. By comparing the classification accuracy of different models for the overall accuracy (OA), we are able to comprehensively assess the performance disparities among the models.

TABLE II
DETAILED NUMBERS OF TRAINING AND TESTING SETS ON TRAINING SCENARIO I

Class	Training Set ($14^\circ - 17^\circ$) Simulated	Training Set ($14^\circ - 17^\circ$) Measured	Testing Set ($14^\circ - 17^\circ$) Measured
2S1	34	34	140
BMP2	21	21	86
BTR70	18	18	69
M1	25	25	104
M2	25	25	103
M35	25	25	104
M548	25	25	103
M60	25	25	141
T-72	21	21	87
ZSU234	34	34	140
Total	268	268	1077

TABLE III
DETAILED NUMBERS OF TRAINING AND TESTING SETS ON TRAINING SCENARIO II

Class	Training Set ($14^\circ - 16^\circ$) Simulated	Training Set ($14^\circ - 16^\circ$) Measured	Testing Set (17°) Measured
2S1	116	116	58
BMP2	55	55	52
BTR70	43	43	49
M1	78	78	51
M2	75	75	53
M35	76	76	53
M548	75	75	53
M60	116	116	60
T-72	56	56	52
ZSU234	116	116	58
Total	806	806	539

Table IV presents the classification performance of different models on the testing set. To mitigate the randomness of individual experiments, we conduct five repeated random samplings of all data in accordance with specified proportions and performed independent experiments. The mean and variance of recognition accuracy are presented in the final results. According to Baseline I, we test the performance of different models with no simulated data input, that is, using measured data alone for training and testing. This experiment, as a scratch method, reflects the ATR performance of the selected backbone model, with the purpose of demonstrating that the use of auxiliary simulated data during the training proecess can improve the ATR accuracy. The experimental results prove that the using of simulated data have benefits on the accuracy of the model. Then, it demonstrates the substantial improvement in target recognition performance can be achieved through appropriate knowledge transfer techniques.

However, it is frustrating that the experiment results of Baseline II shows that the simple mixture of simulated data and measured data cannot effectively promote the ATR

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TABLE IV

COMPARISON OF RECOGNITION ACCURACY (%) BETWEEN THE PROPOSED METHOD AND DA, KD-BASED METHODS WITH DIFFERENT BACKBONES UNDER TRAINING SCENARIO I. BOLD AND VALUES WITH † INDICATE THE BEST AND SECOND BEST ACCURACY, RESPECTIVELY

Backbone		VGG16	ResNet18	ResNet34	SENet	ResNeXt34
Baseline I (Scratch)		89.61±1.39	92.37±1.16	93.75±1.29	94.02±0.51	93.15±0.36
Baseline II (Mix)		88.94±0.24	91.67±1.07	92.27±1.95	93.85±0.92	92.38±0.41
DA	DAN	90.67±0.12	89.55±1.55	93.94±1.71	94.60±0.83	91.95±0.74
	DANN	91.36±0.81	91.51±0.74	91.70±1.08	92.44±1.34	91.73±0.28
	CDAN	88.03±1.53	91.33±0.37	92.91±0.72	93.66±1.09	92.17±1.15
	Deep Coral	89.07±1.14	92.72±1.62	95.90±1.34†	92.82±0.83	92.27±0.65
	DJDAN	92.06±0.83	91.41±1.22	92.90±1.05	93.67±0.97	92.04±0.28
KD	Vanilla KD	90.22±0.74	93.08±0.78	94.08±0.96	96.07±0.75	93.52±0.83
	AB	90.63±0.92	94.59±1.04	95.67±0.48	96.39±0.28†	91.98±1.24
	FitNet	91.27±0.34	95.15±0.85	93.47±0.81	94.13±0.74	93.81±1.71
	NST	92.04±1.28	95.80±0.38†	94.03±1.15	96.36±0.18	92.98±1.26
	SP	91.88±0.31	95.62±0.69	93.48±0.92	94.15±0.67	92.86±1.56
	PKT	92.53±1.08†	93.56±1.03	94.63±1.25	95.71±0.28	94.26±0.57†
Ours		**93.12±1.47**	**96.08±1.22**	**96.94±1.43**	**97.04±1.15**	**94.50±0.63**

accuracy. For all the alternative backbones, without exception, Baseline IIP performed worse than Baseline I. This phenomenon is due to the difference between the simulated and measured data, the introduction of out of distribution information damages the generalization of the model and reduces the accuracy. Similar experimental results have been noted and analyzed in previous studies. For instance, according to Inkawhich et al. [27], with the reduction of the proportion of measured data in the training set, the test accuracy decreases rapidly.

To further explain the distributional differences between the two types of data in the SAMPLE dataset, we conducted data visualization experiments using the t-SNE method. This helped explain why simple mixing the simulated and measured data does not lead to performance improvement. As shown in Fig. 5, after using t-SNE to obtain low-dimensional representations of simulated and measured data, it is obvious that the coincidence degree of their point clusters is low, indicating that there is a significant difference in the distribution of the above two types of data. Thus, for SAR-ATR on the measured data, the simulated SAR images are out of distribution information, which will reduce the performance of the model without taking appropriate alignment methods.

When it comes to the comparison of our proposed method and traditional DA and KD methods, experimental results also fully demonstrate the excellent performance of our proposed model. First, as to the DA method, the accuracy is generally slightly higher or close to Baseline I and Baseline II. For an instance, take ResNet34 as the backbone, the average accuracy of Baseline I and Baseline II is 93.75% and 92.27%, the average accuracy of DAN, DANN, CDAN, Deep Coral, and DJDAN is 93.94%, 91.70%, 92.91%, 95.90%, and 92.90%. It is worth noting that KD-based methods often achieve higher accuracy than DA-based methods. For the experiments taking ResNet34 as the backbone, the average accuracy of Vanilla KD, AB, FitNet, NST, and PKT is 94.08%, 95.67%, 93.47%, 94.03%, 93.48%, and 94.63%. The reason may be that due to the difference between simulated data and measured data, the marginal distribution cannot be effectively aligned by the same feature encoder.

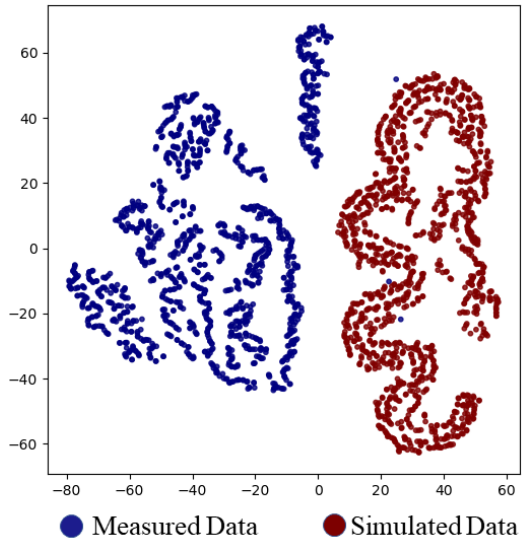


Fig. 5. t-SNE visualization results of the measured and simulated data from the SAMPLE dataset.

Most important of all, the experimental results show that the proposed method demonstrates the optimal performance under all situations of different backbones. In addition, with the enhancement of feature extraction capability of backbone, the accuracy of our proposed method can be further improved.

2) Experiment Results Under Training Scenario II: The purpose of Training Scenario II is to demonstrate the performance of the proposed method in the face of difference in depression angle for simulated and measured data. As shown

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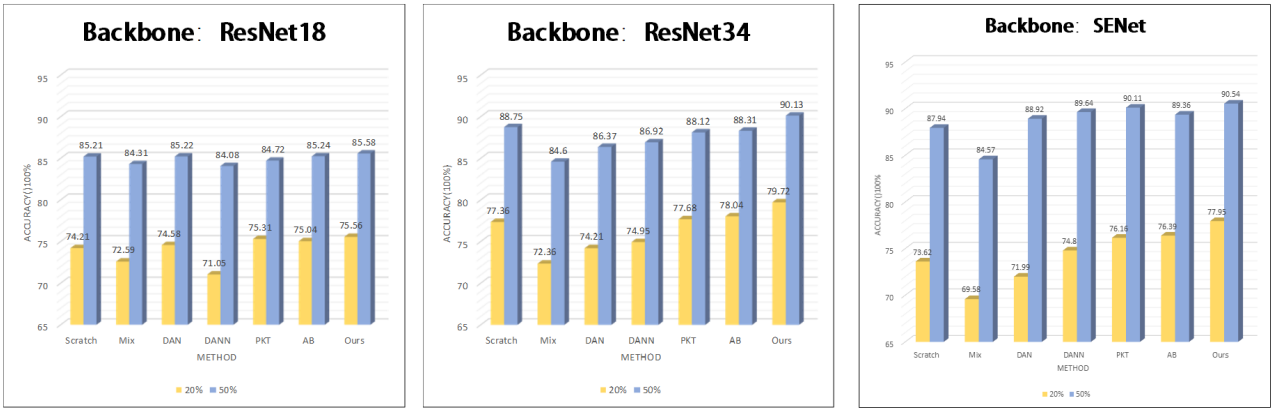


Fig. 6. Accuracy comparison of different methods taking different backbones under Training Scenario II. (a) Results when using ResNet18 as the backbone. (b) Results when using ResNet34 as the backbone. (c) Results when using SENet as the backbone.

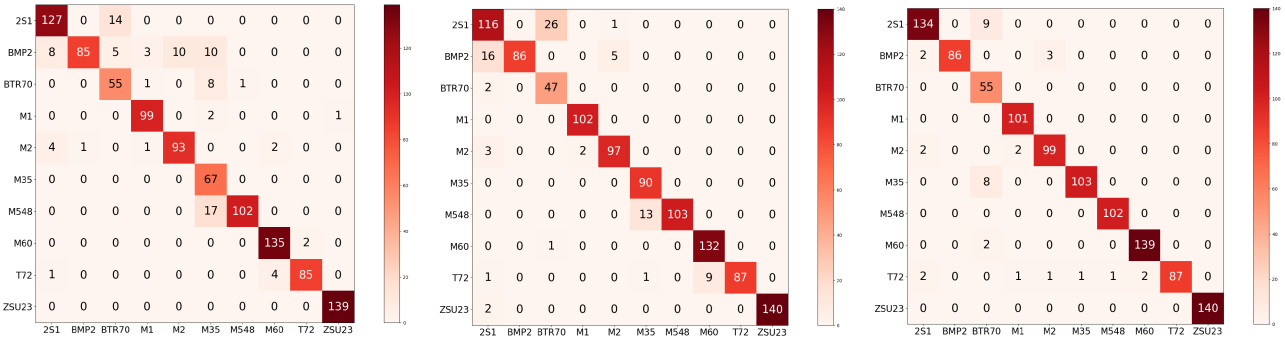


Fig. 7. Comparison of recognition accuracy (%) of different models based on ResNet18 backbone. (a) Results of ResNet18 (OA: 91.22%). (b) Results of ResNet18 with the FSEM (OA: 92.42%). (c) Results of ResNet18 with our proposed method (OA: 96.67%).

in Fig. 6, based on three mainstream backbone networks, we compare the performance difference between the proposed method and various DA and KD methods. The experimental results are roughly consistent with Training Scenario I: No matter taking which backbones, our proposed method achieves the highest performance in all experiment settings. When select 20% sample pairs from the training set for training, as for the ResNet18, ResNet34, and SENet, the proposed method obtained the ATR accuracy of 75.56%, 79.72%, and 77.95%, respectively. While select 50%, the ATR accuracy of our method is 85.58%, 90.13%, and 90.54%, respectively.

As shown in Fig. 6, KD-based methods perform better than the DA-based method when the depression angle of simulated and measured data is different. Similar to the results of Training Scenario I, it proves that KD can effectively avoid the overfitting of the encoder network to the data in a certain domain, and obtain better results when the data in different domains have great difference. Thus, the KD-based methods frequently show better robustness when using simulated SAR data as auxiliary information to boost the ATR performance of measured SAR data. However, we have to point out that this trend is not absolute and will be affected by many variables, including backbone networks and specific knowledge transfer methods.

D. Ablation Studies

Due to the inclusion of both the FSEM and CCM modules in the proposed KD model, along with variations in different hyperparameters, it is imperative to conduct comprehensive ablation experiments.

1) Analysis of FSEM and CCM: Tables V and VI display the classification accuracy of the model for each of the ten target categories in the SAMPLE dataset when VGG16 and ResNet18 are employed as the backbones. When VGG16 and ResNet18 are used as the backbone, the experimental results indicate that our proposed method enhances the accuracy for six and seven out of the ten target categories. Furthermore, Fig. 7 depicts the disparities in the classification confusion matrix among the three models based on the ResNet18 backbone. The visualization of these results provides an intuitive understanding and validation of the effectiveness and robustness of our proposed method. Overall, the experimental data, along with the analysis and visualization, confirm the efficacy and reliability of our proposed method.

When using VGG16 without considering the assistance of simulated data, the accuracy on the testing set is 88.37%. However, upon introducing simulated data and conducting equal-weight KD through the FSEM, the accuracy improves to 91.07%, albeit only slightly. Finally, by employing our proposed method, which incorporates both the CCM and FSEM, the accuracy of VGG16 experiences a significant enhancement, reaching 94.33%.

Similar patterns are observed when ResNet18 is employed as the backbone. Without the utilization of simulated data, the accuracy on the testing set is 91.22%. Upon incorporating simulated data with equal-weight through KD, the accuracy

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TABLE V
COMPARISON OF RECOGNITION ACCURACY (%) WITH VGG16

Class	VGG16	VGG16+FSEM	VGG16+Ours
2S1	81.40	85.33	92.52
BMP2	51.86	61.48	85.00
BTR70	89.23	91.37	100.00
M1	98.91	98.00	100.00
M2	96.47	95.90	100.00
M35	98.67	100.00	100.00
M548	100.00	97.87	78.46
M60	100.00	98.08	95.89
T72	91.40	97.78	98.80
ZSU23	97.01	94.20	100.00
Total	88.37	91.07	94.33

TABLE VI
COMPARISON OF RECOGNITION ACCURACY (%) WITH RESNET18

Class	ResNet18	ResNet18+FSEM	ResNet18+Ours
2S1	90.07	81.12	93.71
BMP2	70.25	80.37	94.51
BTR70	84.62	95.92	100.00
M1	97.06	100.00	100.00
M2	92.08	95.11	96.12
M35	100.00	100.00	92.80
M548	85.71	88.79	100.00
M60	98.54	99.25	98.58
T72	94.44	88.78	91.58
ZSU23	100.00	98.61	100.00
Total	91.22	92.42	96.67

increases to 92.42%. However, the expected results are not achieved by transferring dark knowledge alone. Nevertheless, after selectively transferring dark knowledge using the CCM, there is a significant improvement in accuracy, reaching 96.67%. These experimental outcomes provide evidence that our proposed method excels at transferring high-quality dark knowledge compared to traditional KD approaches. As a result, it enables the effective utilization of simulated data to enhance SAR-ATR tasks involving measured data.

2) Analysis of Hyperparameters: Analyzing the influence of hyperparameters on SAR-ATR performance is an important step in understanding the behavior of our proposed method. In this analysis, we focus on two key hyperparameters: λ_1 and λ_2. These hyperparameters control the weights of the loss functions $\mathcal{L}_{\text{CCM}}$ and $\mathcal{L}_{\text{FSEM}}$ associated with the CCM and the FSEM, respectively. To investigate the influence of these hyperparameters, we utilize ResNet18 as the backbone and incorporate the CCM and FSEM into the model. To explore the effects of both hyperparameters simultaneously, we employ the grid search method to conduct the following experiments. The grid search method involves systematically varying the values of λ_1 and λ_2 within a predefined range and evaluating the resulting SAR-ATR performance. By examining the performance across different combinations of hyperparameter values, we can gain insights into the optimal settings for λ_1 and λ_2 and their impact on the overall performance of the proposed method. Through this detailed analysis, we aim to identify the hyperparameter configurations that yield the best SAR-ATR performance, providing valuable guidance for practical implementation and fine-tuning of our proposed method.

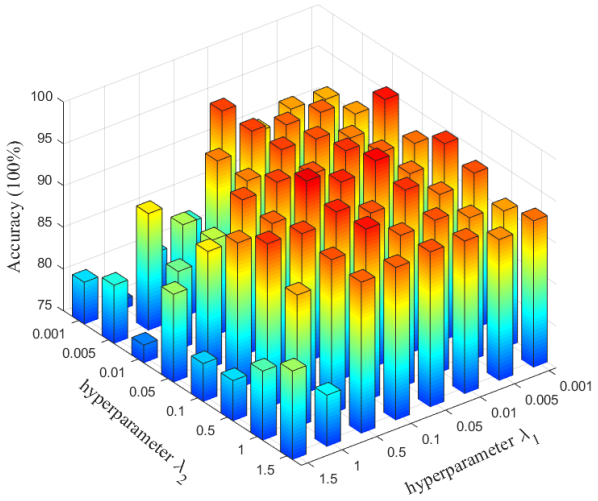


Fig. 8. Effect of hyperparameters λ_1 and λ_2 on the recognition accuracy.

We set the values of λ_1 and λ_2 to the numbers in the set {0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 1.5}, which allows for a comprehensive exploration of different hyperparameter combinations. The chosen values of λ_1 and λ_2 span a range of magnitudes, from small values such as 0.001 and 0.005 to larger values like 1.0 and 1.5. This selection allows for a thorough investigation of the effects of the hyperparameters and provides insights into the optimal balance between the contributions of the CCM and FSEM. The results obtained from the experiments conducted are summarized in Fig. 8.

The experimental results indicate that the choice of hyperparameter values has a substantial influence on the target recognition accuracy. Among the tested combinations, the model achieves the highest accuracy when both λ_1 and λ_2 are set to 0.1. It is important to note that any deviation from this specific combination, whether by increasing or decreasing the values of λ_1 and λ_2, leads to a degradation in the performance of the model. This finding suggests that maintaining a balance between the contributions of the CCM and FSEM is critical for achieving optimal results.

V. CONCLUSION

Different from the common practice of using simulated data to improve the performance on measured data in SAR-ATR, in this article, we present a detailed mathematical derivation for estimating the validity of dark knowledge extracted from simulated data, which underpins our introduction of the concept of trusted simulated-measured KD. Specifically, we model the distribution of predicted probabilities with an evidence-parameterized Dirichlet distribution based on the pipeline of evidential learning. This enables the computation of the belief mass and uncertainty associated with each simulated sample. Moreover, we introduce a novel distillation objective function that maximizes the shared semantic information in paired simulated-measured feature maps in a trusted manner, so as to enhance the distillation performance while ensuring

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plug-and-play. The proposed method significantly improves the reliability and robustness of performing KD from simulated data, and can be seamlessly incorporated into any DL-based SAR-ATR model. Experiments on the SAMPLE dataset verified the superior performance of our proposed method over comparing methods.

Moving forward, we will explore the potential of trusted KD in more SAR application fields, such as target detection and image segmentation. We will also explore trusted DL with other learning pipelines, such as self-supervised learning, few-shot learning, model compression, and so on.

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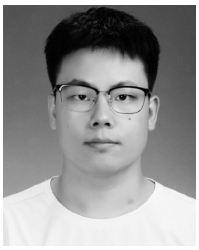
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